Practice Exam 2

MTH 207 (Fall 2026) — J. Baggett

Score

/ 134

This is practice, not a preview. The real exam will **not** be these problems with the numbers changed. Expect different functions, and expect at least one technique this sheet happens not to show. Everything on the review list is fair game whether or not it turns up here — use this to find out what you cannot yet do, not to predict what will be asked.

This sheet is also deliberately **longer** than the real exam (134 points against 100) so that it covers more ground, so do not use it to judge timing. The real exam is 55 minutes.

First Name	Last Name

Before exam begins:

- Write your first name and last name above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 7 pages of the exam. (The number of pages includes this cover page.)

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right] = \dfrac{f(x)g'(x) - f'(x)g(x)}{[g(x)]^2}$.

(b) If $h(x) = f(g(x))$ with $g(1) = 1$, $g'(1) = 3$ and $f'(1) = 5$, then $h'(1) = 15$.

(c) $\ln(x^2) = 2\ln x$ for every $x \neq 0$.

(d) If f is differentiable and $f(-1) = f(1)$, then there is a number c with $|c| < 1$ and $f'(c) = 0$.

/12

(6^{pts}) **2.** Evaluate the limit with **justifications**. Simplify until you have a number.

$\lim\limits_{x\to 0} \dfrac{\tan 4x}{\sin 7x}$

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(6^{pts}) **3.** A table of values for f, g, f' and g' is given. If $h(x) = f(g(x))$, find $h'(1)$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	4	6
2	1	8	5	7
3	7	2	7	9

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(6^{pts}) **4.** Compute the linearization of $f(x) = x^3$ at the x-value 2.

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(6^{pts}) **5.** Find the 50th derivative of $y = \cos 2x$.

(9^{pts}) **6.** If y depends on x according to $\cos(xy) = 1 + \sin y$, find $\dfrac{dy}{dx}$.

/9

(7^{pts} ea.) **7.** Derive the following derivative rules:

(a) Derive the derivative rule for $y = \tan x$. (You may use the derivative rules for sine and cosine.) This question is about **regular tangent**, not inverse tangent.

(b) Derive the derivative rule for $y = \sin^{-1} x$. (You may use derivative rules for the trig functions, but not any derivative rules for inverse trig functions.) This question is about **inverse sine**.

(c) Given a fixed number $a > 0$, derive the derivative rule for $y = \log_a(x)$. (You may use the derivative rule for a^x.)

/21

(5^{pts} ea.) **8.** Find the derivative of each function below. (Use the derivative rules from the **previous page**!)

(a) $f(x) = 3x^{10} - x^5 + 2x^3 + \sin(x)$

(b) $y = x^{\sin x}$

(c) $f(x) = \Big(\sin x\Big)\Big(\log_4 x\Big) + \ln(x)$

(d) $y = \cos\left(\dfrac{\sin^{-1}(x)}{3x^4}\right) + \sin(5^{\tan x})$

/20

(20^{pts}) **9.** A 13 ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec. How fast is the top of the ladder sliding down the wall then? (Be sure to include units.)

/20

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(8pts) **10.** Show that the equation $x^3 + e^x = 0$ has exactly one real root.

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Not on Exam 2. This one uses an argument we take up *after* the exam, so do not let it stall your review this weekend. The worked solution is worth reading once you have done the rest — it is two separate ideas (something exists, and there is only one of it) and that pairing comes back.

(12pts) **11.** Let $f(x) = xe^x - e^x$. Find the absolute maximum point and absolute minimum point on the interval $[-1, 1]$. (Hint: Use the fact that $e > 2$.)

/12

(8^{pts}) **12.** Verify that $f(x) = 1/x$ on $[1, 3]$ satisfies the hypotheses of the Mean Value Theorem (clearly state why each hypothesis is satisfied) and find all numbers c that satisfy the conclusion of the Mean Value Theorem.

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